

EXT 1: Trigonometry (Ext1)

T2 Further Trigonometric Identities (Y11)

EXT 1: Trigonometry (Ext1), T3 Trig Equations (Y12)

Auxiliary Angles (Ext1)

Identities, Equations and 't' formulae (Ext1)

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Exam Equivalent Time: 171 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Trigonometry, EXT1 T3 2014 HSC 2 MC

Which expression is equal to $\cos x - \sin x$?

- (A) $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$
- (B) $\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$
- (C) $2 \cos\left(x + \frac{\pi}{4}\right)$
- (D) $2 \cos\left(x - \frac{\pi}{4}\right)$

2. Trigonometry, EXT1 T2 2016 HSC 3 MC

Which expression is equivalent to $\frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$?

- (A) $\tan x$
- (B) $\tan 3x$
- (C) $\frac{\tan 2x - 1}{1 + \tan 2x}$
- (D) $\frac{\tan x}{1 + \tan 2x \tan x}$

3. Trigonometry, EXT1 T2 SM-Bank 9 MC

If $\sin(\theta + \phi) = a$ and $\sin(\theta - \phi) = b$, then $\sin(\theta)\cos(\phi)$ is equal to

- A. $\sqrt{a^2 + b^2}$
- B. $\sqrt{ab}$
- C. $\sqrt{a^2 - b^2}$
- D. $\frac{a + b}{2}$

4. Trigonometry, EXT1 T3 2017 HSC 4 MC

What is the value of $\tan \alpha$ when the expression $2 \sin x - \cos x$ is written in the form

$\sqrt{5} \sin(x - \alpha)$?

- A. -2
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 2

5. Trigonometry, EXT1 T2 2019 HSC 6 MC

It is given that $\sin x = \frac{1}{4}$, where $\frac{\pi}{2} < x < \pi$.

What is the value of $\sin 2x$?

- A. $-\frac{7}{8}$
- B. $-\frac{\sqrt{15}}{8}$
- C. $\frac{\sqrt{15}}{8}$
- D. $\frac{7}{8}$

6. Trigonometry, EXT1 T2 2013 HSC 8 MC

The angle θ satisfies $\sin \theta = \frac{5}{13}$ and $\frac{\pi}{2} < \theta < \pi$.

What is the value of $\sin 2\theta$?

- (A) $\frac{10}{13}$
 (B) $-\frac{10}{13}$
 (C) $\frac{120}{169}$
 (D) $-\frac{120}{169}$

7. Trigonometry, EXT1 T2 SM-Bank 3

Show that

$$\sin(8x + 3x) + \sin(8x - 3x) = 2 \sin 8x \cos 3x. \quad (1 \text{ mark})$$

8. Trigonometry, EXT1 T2 2006 HSC 1d

i. Simplify $(\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)$ (1 mark)

ii. Hence, or otherwise, express $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} - 1$, in its simplest form for $0 < \theta < \frac{\pi}{2}$. (2 marks)

9. Trigonometry, EXT1 T3 2015 HSC 11d

Express $5 \cos x - 12 \sin x$ in the form $A \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)

10. Trigonometry, EXT1 T3 2018 HSC 11c

Write $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)

11. Trigonometry, EXT1 T2 EQ-Bank 1

Find a and b such that

$$\tan 75^\circ = a + b\sqrt{3} \quad (2 \text{ marks})$$

12. Trigonometry, EXT1 T2 SM-Bank 4

If $\cos \theta = -\frac{3}{4}$ and $0 < \theta < \pi$,

determine the exact value of $\tan \theta$. (2 marks)

13. Trigonometry, EXT1 T2 SM-Bank 6

Find the exact value of $\sin \frac{\pi}{12}$. (2 marks)

14. Trigonometry, EXT1 T3 SM-Bank 10

Given that $\cot(2x) + \frac{1}{2} \tan(x) = a \cot(x)$, calculate a . (3 marks)

15. Calculus, EXT1 C2 2005 HSC 3b

i. By expanding the left-hand side, show that

$$\sin(5x + 4x) + \sin(5x - 4x) = 2 \sin 5x \cos 4x \quad (1 \text{ mark})$$

ii. Hence find $\int \sin 5x \cos 4x \, dx$. (2 marks)

16. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3} \cos x - \sin x$ in the form $2 \cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. (1 mark)

ii. Hence, or otherwise, solve $\sqrt{3} \cos x = 1 + \sin x$, where $0 < x < 2\pi$. (2 marks)

17. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8 \cos x + 6 \sin x$ in the form $A \cos(x - \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)

ii. Hence, or otherwise, solve the equation $8 \cos x + 6 \sin x = 5$ for $0 \leq x \leq 2\pi$.
 Give your answers correct to three decimal places. (2 marks)

18. Trigonometry, EXT1 T3 2007 HSC 2a

By using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$. (2 marks)

19. Trigonometry, EXT1 T3 2009 HSC 2b

- i. Express $3 \sin x + 4 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)
- ii. Hence, or otherwise, solve $3 \sin x + 4 \cos x = 5$ for $0 \leq x \leq 2\pi$.
Give your answer, or answers, correct to two decimal places. (2 marks)

20. Trigonometry, EXT1 T3 2009 HSC 3c

- i. Prove that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ provided that $\cos 2\theta \neq -1$. (2 marks)
- ii. Hence find the exact value of $\tan \frac{\pi}{8}$. (1 mark)

21. Trigonometry, EXT1 T3 2010 HSC 6a

- i. Show that $\cos(A - B) = \cos A \cos B(1 + \tan A \tan B)$. (1 mark)
- ii. Suppose that $0 < B < \frac{\pi}{2}$ and $B < A < \pi$.
Deduce that if $\tan A \tan B = -1$, then $A - B = \frac{\pi}{2}$. (1 mark)

22. Trigonometry, EXT1 T3 SM-Bank 13

Find all solutions of $\tan(2\theta) = -\tan \theta$ for $0 \leq \theta \leq 2\pi$. (3 marks)

23. Trigonometry, EXT1 T2 SM-Bank 2

Find the exact value of $\cos \frac{\pi}{8}$. (2 marks)

24. Trigonometry, EXT1 T2 EQ-Bank 5

If $\sin \theta = -\frac{4}{6}$ and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$,

determine the exact value of $\cos \theta$ in its simplest form. (2 marks)

25. Trigonometry, EXT1 T2 EQ-Bank 7

Find the exact value of $\cos\left(\frac{11\pi}{12}\right)$. (2 marks)

26. Trigonometry, EXT1 T3 SM-Bank 11

Given that $\cos(\theta - \phi) = \frac{3}{5}$ and $\tan \theta \tan \phi = 2$, find $\cos(\theta + \phi)$. (3 marks)

27. Trigonometry, EXT1 T3 SM-Bank 12

Solve $\sin(2x) = \sin x$, for $0 \leq x \leq 2\pi$. (3 marks)

28. Trigonometry, EXT1 T3 SM-Bank 8

Solve for x , given that

$$x \sin x \sec(2x) = 0, \quad 0 \leq x \leq 2\pi \quad (2 \text{ marks})$$

29. Trigonometry, EXT1 T3 2020 HSC 11d

By expressing $\sqrt{3} \sin x + 3 \cos x$ in the form $A \sin(x + \alpha)$, solve $\sqrt{3} \sin x + 3 \cos x = \sqrt{3}$, for $0 \leq x \leq 2\pi$. (4 marks)

30. Trigonometry, EXT1 T3 2021 HSC 11g

By factorising, or otherwise, solve $2 \sin^3 x + 2 \sin^2 x - \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. (3 marks)

31. Trigonometry, EXT1 T3 EQ-Bank 3

Find all values of θ that satisfy the equation $\sqrt{3} \cos \theta = \sin(2\theta)$. (3 marks)

32. Trigonometry, EXT1 T3 EQ-Bank 4

The current flowing through an electrical circuit can be modelled by the function

$$f(t) = 6 \sin 0.05t + 8 \cos 0.05t, \quad t \geq 0$$

- i. Express the function in the form $f(t) = A \sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$. (2 marks)
- ii. Find the time at which the current first obtains its maximum value. (1 mark)
- iii. Sketch the graph of $f(t)$. Clearly show its range and label the coordinates of its first maximum value. Do not label x -intercepts. (1 mark)

33. Trigonometry, EXT1 T3 SM-Bank 2

Show that

$$\cos 3x = 4 \cos^3 x - 3 \cos x. \quad (3 \text{ marks})$$

34. Trigonometry, EXT1 T2 2021 HSC 13d

- i. The numbers A , B and C are related by the equations $A = B - d$ and $C = B + d$, where d is a constant.

Show that $\frac{\sin A + \sin C}{\cos A + \cos C} = \tan B$. (2 marks)

- ii. Hence, or otherwise, solve $\frac{\sin \frac{5\theta}{7} + \sin \frac{6\theta}{7}}{\cos \frac{5\theta}{7} + \cos \frac{6\theta}{7}} = \sqrt{3}$ for $0 \leq \theta \leq 2\pi$. (2 marks)

35. Trigonometry, EXT1 T3 2010 HSC 4b

- i. Express $2 \cos \theta + 2 \cos\left(\theta + \frac{\pi}{3}\right)$ in the form $R \cos(\theta + \alpha)$,

where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. (3 marks)

- ii. Hence, or otherwise, solve $2 \cos \theta + 2 \cos\left(\theta + \frac{\pi}{3}\right) = 3$,

for $0 < \theta < 2\pi$. (2 marks)

36. Trigonometry, EXT1 T3 EQ-Bank 1

- i. Show that $\sin x + \sin 3x = 2 \sin 2x \cos x$. (2 marks)

- ii. Hence or otherwise, find all values of x that satisfy

$$\sin x + \sin 2x + \sin 3x = 0, \quad x \in [0, 2\pi]. \quad (2 \text{ marks})$$

37. Trigonometry, EXT1 T3 EQ-Bank 2

A particular energy wave can be modelled by the function

$$f(t) = \sqrt{5} \sin 0.2t + 2 \cos 0.2t, \quad t \in [0, 50]$$

- i. Express this function in the form $f(t) = R \sin(nt - \alpha)$, $\alpha \in [0, 2\pi]$. (2 marks)

- ii. Find the time the wave first attains its maximum value. Give your answer to one decimal place. (2 marks)

38. Trigonometry, EXT1 T3 2005 HSC 4b

By making the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that

$$\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}. \quad (2 \text{ marks})$$

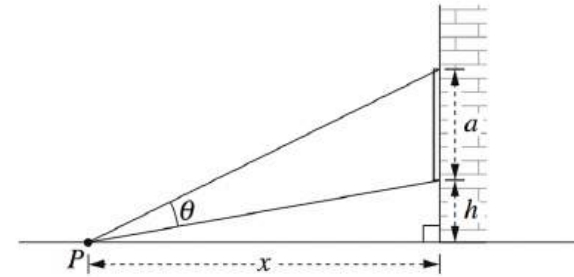
39. Trigonometry, EXT1 T3 2008 HSC 6b

It can be shown that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ for all values of θ . (Do NOT prove this.)

Use this result to solve $\sin 3\theta + \sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$. (3 marks)

40. Trigonometry, EXT1 T3 SM-Bank 1

A billboard of height a metres is mounted on the side of a building, with its bottom edge h metres above street level. The billboard subtends an angle θ at the point P , x metres from the building.



Use the identity $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ to show that

$$\theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a + h)} \right]. \quad (2 \text{ marks})$$

41. Trigonometry, EXT1 T3 2020 HSC 14b

- i. Show that $\sin^3 \theta - \frac{3}{4} \sin \theta + \frac{\sin(3\theta)}{4} = 0$. (2 marks)

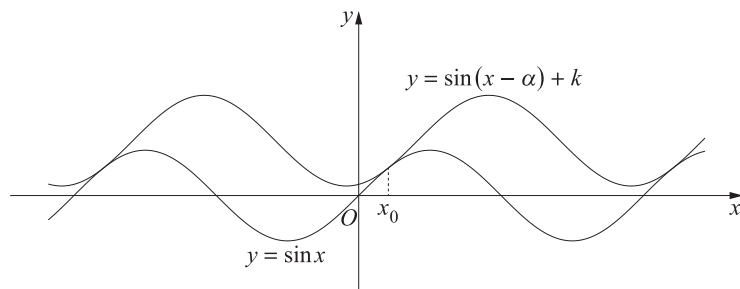
- ii. By letting $x = 4 \sin \theta$ in the cubic equation $x^3 - 12x + 8 = 0$.

Show that $\sin(3\theta) = \frac{1}{2}$. (2 marks)

- iii. Prove that $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$. (3 marks)

42. Calculus, EXT1 C2 2019 HSC 14c

The diagram shows the two curves $y = \sin x$ and $y = \sin(x - \alpha) + k$, where $0 < \alpha < \pi$ and $k > 0$. The two curves have a common tangent at x_0 where $0 < x_0 < \frac{\pi}{2}$.



- Explain why $\cos x_0 = \cos(x_0 - \alpha)$. (1 mark)
- Show that $\sin x_0 = -\sin(x_0 - \alpha)$. (2 marks)
- Hence, or otherwise, find k in terms of α . (2 marks)

Worked Solutions

- Trigonometry, EXT1 T3 2014 HSC 2 MC

$$R \cos(x + \alpha) = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\therefore R \cos x \cos \alpha - R \sin x \sin \alpha = \cos x - \sin x$$

$$R \cos \alpha = 1, \quad R \sin \alpha = 1$$

$$R^2 = 1^2 + 1^2$$

$$R = \sqrt{2}$$

$$\therefore \sqrt{2} \cos \alpha = 1$$

$$\cos \alpha = \frac{1}{\sqrt{2}}$$

$$\alpha = \frac{\pi}{4}$$

$$\therefore \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = \cos x - \sin x$$

$$\Rightarrow A$$

- Trigonometry, EXT1 T2 2016 HSC 3 MC

$$\frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$$

$$= \tan(2x - x)$$

$$= \tan x$$

$$= \tan x$$

$$\Rightarrow A$$

3. Trigonometry, EXT1 T2 SM-Bank 9 MC

Solution 1

$$\sin \theta \cos \phi = \frac{1}{2}[(\sin(\theta + \phi) + \sin(\theta - \phi))]$$

$$= \frac{1}{2}(a + b)$$

$\Rightarrow D$

Solution 2

$$\sin(\theta + \phi) = a$$

$$\sin \theta \cos \phi + \sin \phi \cos \theta = a \dots (1)$$

$$\sin(\theta - \phi) = b$$

$$\sin \theta \cos \phi - \sin \phi \cos \theta = b \dots (2)$$

(1) + (2):

$$2 \sin \theta \cos \phi = a + b$$

$$\therefore \sin \theta \cos \phi = \frac{a + b}{2}$$

4. Trigonometry, EXT1 T3 2017 HSC 4 MC

$$\text{Using } \sqrt{5} \sin(x - \alpha) = \sqrt{5} \sin x \cos \alpha - \sqrt{5} \cos x \sin \alpha,$$

$$\Rightarrow \sqrt{5} \sin x \cos \alpha - \sqrt{5} \cos x \sin \alpha = 2 \sin x - \cos x$$

$$\Rightarrow \sqrt{5} \cos \alpha = 2, \sqrt{5} \sin \alpha = 1$$

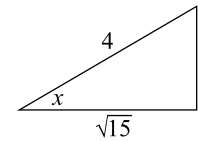
$$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{2}$$

$$\therefore \tan \alpha = \frac{1}{2}$$

$\Rightarrow C$

5. Trigonometry, EXT1 T2 2019 HSC 6 MC

$$\sin x = \frac{1}{4}$$



$$\cos x = -\frac{\sqrt{15}}{4}, \left(\frac{\pi}{2} < x < \pi\right)$$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{1}{4} \times -\frac{\sqrt{15}}{4}$$

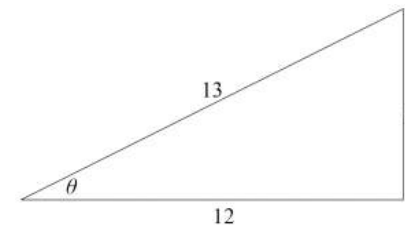
$$= -\frac{\sqrt{15}}{8}$$

$\Rightarrow B$

6. Trigonometry, EXT1 T2 2013 HSC 8 MC

♦ Mean mark 44%

$$\sin \theta = \frac{5}{13} \left(\frac{\pi}{2} < \theta < \pi\right)$$



Since θ is in the 2nd quadrant,

$$\cos \theta = -\frac{12}{13}$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times \frac{5}{13} \times -\frac{12}{13}$$

$$= -\frac{120}{169}$$

$\Rightarrow D$

7. Trigonometry, EXT1 T2 SM-Bank 3

Expanding the LHS:

$$\begin{aligned}\text{LHS} &= \sin 8x \cos 3x + \cos 8x \sin 3x + \sin 8x \cos 3x - \cos 8x \sin 3x \\ &= 2 \sin 8x \cos 3x\end{aligned}$$

8. Trigonometry, EXT1 T2 2006 HSC 1d

$$\begin{aligned}\text{i. } (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta) \\ &= \sin^3 \theta - \sin^2 \theta \cos \theta + \sin \theta \cos^2 \theta + \cos \theta \sin^2 \theta - \sin \theta \cos^2 \theta + \cos^3 \theta \\ &= \sin^3 \theta + \cos^3 \theta\end{aligned}$$

$$\begin{aligned}\text{ii. } \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} - 1 \\ &= \frac{(\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)}{\sin \theta + \cos \theta} - 1 \\ &= \sin^2 \theta + \cos^2 \theta - \sin \theta \cos \theta - 1 \\ &= 1 - \sin \theta \cos \theta - 1 \\ &= -\sin \theta \cos \theta\end{aligned}$$

9. Trigonometry, EXT1 T3 2015 HSC 11d

$$\begin{aligned}5 \cos x - 12 \sin x &= A \cos(x + \alpha) \\ &= A \cos x \cos \alpha - A \sin x \sin \alpha\end{aligned}$$

$$\therefore A \cos \alpha = 5, \quad A \sin \alpha = 12$$

$$A^2 = 5^2 + 12^2 = 169$$

$$A = 13$$

$$\Rightarrow 13 \cos \alpha = 5$$

$$\cos \alpha = \frac{5}{13}$$

$$\alpha = \cos^{-1}\left(\frac{5}{13}\right) \approx 1.176 \dots \text{radians}$$

$$\therefore 5 \cos x - 12 \sin x = 13 \cos(x + 1.176 \dots)$$

10. Trigonometry, EXT1 T3 2018 HSC 11c

$$R \sin(x + \alpha) = \sqrt{3} \sin x + \cos x$$

$$R \sin x \cos \alpha + R \cos x \sin \alpha = \sqrt{3} \sin x + \cos x$$

Equating coefficients:

$$\Rightarrow R \cos \alpha = \sqrt{3}$$

$$\Rightarrow R \sin \alpha = 1$$

$$R^2 = (\sqrt{3})^2 + 1^2 = 4$$

$$\therefore R = 2$$

$$\Rightarrow 2 \sin \alpha = 1$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6} \quad \left(0 \leq \alpha \leq \frac{\pi}{2}\right)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{\pi}{6}\right).$$

11. Trigonometry, EXT1 T2 EQ-Bank 1

$$\tan 75^\circ = \tan(45 + 30)^\circ$$

$$= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \cdot \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

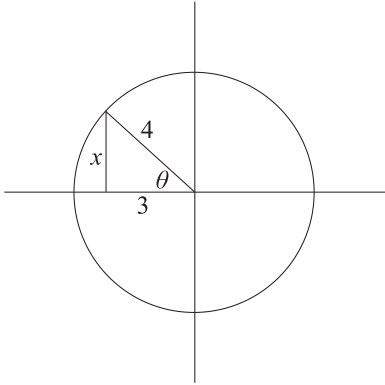
$$= \frac{3 + 2\sqrt{3} + 1}{(\sqrt{3})^2 - 1^2}$$

$$= 2 + \sqrt{3}$$

$$\therefore a = 2, \quad b = 1$$

12. Trigonometry, EXT1 T2 SM-Bank 4

Consider the angle graphically:



Since $\cos \theta$ is negative $\Rightarrow$ 2nd quadrant.

Using Pythagoras,

$$x^2 = 4^2 - 3^2$$

$$x = \sqrt{7}$$

$$\therefore \tan \theta = \frac{\sqrt{7}}{3}$$

13. Trigonometry, EXT1 T2 SM-Bank 6

$$\begin{aligned} \sin \frac{\pi}{12} &= \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= \sin \frac{\pi}{3} \cdot \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \cdot \sin \frac{\pi}{4} \\ &= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{6} - \sqrt{2}}{4} \end{aligned}$$

14. Trigonometry, EXT1 T3 SM-Bank 10

$$\frac{1}{\tan(2x)} + \frac{\tan(x)}{2} = \frac{a}{\tan(x)}, \quad \tan(x) \neq 0$$

$$\frac{1 - \tan^2(x)}{2 \tan(x)} + \frac{\tan(x)}{2} = \frac{a}{\tan(x)}$$

If $\tan(x) \neq 0$:

$$\frac{1 - \tan^2(x) + \tan^2(x)}{2 \tan(x)} = \frac{a}{\tan(x)}$$

$$1 = 2a$$

$$\therefore a = \frac{1}{2}$$

15. Calculus, EXT1 C2 2005 HSC 3b

i. $\sin(5x + 4x) + \sin(5x - 4x) = 2 \sin 5x \cos 4x$

$$\begin{aligned} \text{LHS} &= \sin 5x \cos 4x - \sin 4x \cos 5x + \sin 5x \cos 4x \\ &\quad + \sin 4x \cos 5x \\ &= 2 \sin 5x \cos 4x \quad \dots \text{ as required} \end{aligned}$$

ii. $\int \sin 5x \cos 4x \, dx$

$$\begin{aligned} &= \frac{1}{2} \int 2 \sin 5x \cos 4x \, dx \\ &= \frac{1}{2} \int \sin(5x + 4x) + \sin(5x - 4x) \, dx \\ &= \frac{1}{2} \int \sin 9x + \sin x \, dx \\ &= \frac{1}{2} \left[-\frac{1}{9} \cos 9x - \cos x \right] + c \\ &= -\frac{1}{18} \cos 9x - \frac{1}{2} \cos x + c \end{aligned}$$

16. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3} \cos x - \sin x$ in form

$$2 \cos(x + \alpha), \quad 0 < \alpha < \frac{\pi}{2}$$

$$2(\cos x \cos \alpha - \sin x \sin \alpha) = \sqrt{3} \cos x - \sin x$$

$$\cos x \cos \alpha - \sin x \sin \alpha = \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x$$

$$\Rightarrow \cos \alpha = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore 2 \cos\left(x + \frac{\pi}{6}\right) = \sqrt{3} \cos x - \sin x$$

ii. Solve $\sqrt{3} \cos x = 1 + \sin x$, $0 < x < 2\pi$

$$\sqrt{3} \cos x - \sin x = 1$$

$$2 \cos\left(x + \frac{\pi}{6}\right) = 1 \quad (\text{from part (i)})$$

$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Since cos is positive in 1st / 4th quadrants

$$x + \frac{\pi}{6} = \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \quad (0 < x < 2\pi)$$

$$\therefore x = \frac{\pi}{6}, \frac{3\pi}{2}$$

17. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8 \cos x + 6 \sin x$ in the form $A \cos(x - \alpha)$

$$A(\cos x \cos \alpha + \sin x \sin \alpha) = 8 \cos x + 6 \sin x$$

$$(\cos x \cos \alpha + \sin x \sin \alpha) = \frac{8}{A} \cos x + \frac{6}{A} \sin x$$

$$\Rightarrow \cos \alpha = \frac{8}{A}$$

$$\Rightarrow \sin \alpha = \frac{6}{A}$$

$$\left(\frac{8}{A}\right)^2 + \left(\frac{6}{A}\right)^2 = 1$$

$$8^2 + 6^2 = A^2$$

$$\therefore A = \sqrt{100}$$

$$= 10$$

$$\Rightarrow \cos \alpha = \frac{8}{10}$$

$$\therefore \alpha = 0.6435 \dots \text{radians}$$

$$\therefore 8 \cos x + 6 \sin x = 10 \cos(x - 0.6435 \dots)$$

ii. $8 \cos x + 6 \sin x = 5$

$$10 \cos(x - 0.6435 \dots) = 5$$

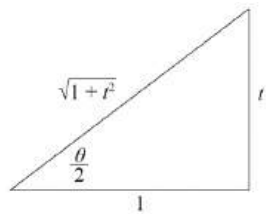
$$\cos(x - 0.6435 \dots) = \frac{1}{2}$$

$$x - 0.6435 \dots = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$x = \frac{\pi}{3} + 0.6435 \dots \text{ or } \frac{5\pi}{3} + 0.6435 \dots$$

$$= 1.691 \text{ or } 5.879 \text{ radians (to 3 d.p.)}$$

18. Trigonometry, EXT1 T3 2007 HSC 2a



COMMENT: The values of $\sin \theta$ and $\cos \theta$ can be committed to memory or quickly derived using double angle formulae (as shown in the worked solution).

$$\Rightarrow \tan \frac{\theta}{2} = t$$

$$\Rightarrow \sin \theta = 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2}$$

$$\Rightarrow \cos \theta = \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$$

Show $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$:

$$\begin{aligned} \frac{1 - \cos \theta}{\sin \theta} &= \frac{1 - \left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}} \times \frac{1+t^2}{1+t^2} \\ &= \frac{1+t^2 - (1-t^2)}{2t} \\ &= \frac{2t^2}{2t} \\ &= t \\ &= \tan \frac{\theta}{2} \dots \text{as required} \end{aligned}$$

19. Trigonometry, EXT1 T3 2009 HSC 2b

i. $A \sin(x + \alpha) = 3 \sin x + 4 \cos x$

$$A \sin x \cos \alpha + A \cos x \sin \alpha = 3 \sin x + 4 \cos x$$

$$\Rightarrow A \cos \alpha = 3 \quad A \sin \alpha = 4$$

$$A^2 = 3^2 + 4^2 = 25$$

$$\therefore A = 5$$

$$\Rightarrow 5 \cos \alpha = 3$$

$$\cos \alpha = \frac{3}{5}$$

$$\alpha = \cos^{-1}\left(\frac{3}{5}\right) = 0.9272\dots \text{ radians}$$

$$\therefore 3 \sin x + 4 \cos x = 5 \sin\left(x + \cos^{-1}\left(\frac{3}{5}\right)\right)$$

ii. $3 \sin x + 4 \cos x = 5$

$$5 \sin(x + \alpha) = 5$$

$$\sin(x + \alpha) = 1$$

$$x + \alpha = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$$

$$x = \frac{\pi}{2} - 0.9272\dots \quad (0 \leq x \leq 2\pi)$$

$$= 0.6435\dots$$

$$= 0.64 \text{ radians (2 d.p.)}$$

20. Trigonometry, EXT1 T3 2009 HSC 3c

i. Prove $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$, $\cos 2\theta \neq -1$

$$\begin{aligned}\text{Using } \cos 2\theta &= 2\cos^2 \theta - 1 \\ &= 1 - 2\sin^2 \theta\end{aligned}$$

$$\begin{aligned}\text{RHS} &= \frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)} \\ &= \frac{2\sin^2 \theta}{2\cos^2 \theta} \\ &= \tan^2 \theta \dots \text{as required.}\end{aligned}$$

ii. $\tan^2 \frac{\pi}{8} = \frac{1 - \cos(2 \times \frac{\pi}{8})}{1 + \cos(2 \times \frac{\pi}{8})}$

$$\begin{aligned}&= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} \\ &= \frac{(\sqrt{2} - 1)^2}{2 - 1} \\ &= (\sqrt{2} - 1)^2\end{aligned}$$

$$\therefore \tan \frac{\pi}{8} = \sqrt{2} - 1 \quad \left(\tan\left(\frac{\pi}{8}\right) > 0 \right)$$

$$\left(\tan \frac{\pi}{8} = \sqrt{3} - 2\sqrt{2} \text{ is also a correct answer} \right)$$

21. Trigonometry, EXT1 T3 2010 HSC 6a

i. Show $\cos(A - B) = \cos A \cos B(1 + \tan A \tan B)$

$$\begin{aligned}\text{RHS} &= \cos A \cos B \left(1 + \frac{\sin A \sin B}{\cos A \cos B} \right) \\ &= \cos A \cos B + \sin A \sin B \\ &= \cos(A - B) \dots \text{as required}\end{aligned}$$

ii. Given $\tan A \tan B = -1$

$$\cos(A - B) = \cos A \cos B(1 - 1)$$

$$\cos(A - B) = 0$$

$$A - B = \cos^{-1} 0$$

$$= \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\text{Since } 0 < B < \frac{\pi}{2} \text{ and } B < A < \pi,$$

$$\Rightarrow A - B = \frac{\pi}{2}$$

22. Trigonometry, EXT1 T3 SM-Bank 13

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = -\tan \theta$$

Let $\tan \theta = k$,

$$\frac{2k}{1 - k^2} = -k$$

$$2k = -k(1 - k^2)$$

$$2k + k(1 - k^2) = 0$$

$$3k - k^3 = 0$$

$$k(3 - k^2) = 0$$

$$k(3 - k^2) = 0 \Rightarrow k = 0, k = \pm \sqrt{3}$$

If $\tan \theta = 0 \Rightarrow \theta = 0, \pi, 2\pi$

If $\tan \theta = \pm \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

$$\therefore \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi$$

23. Trigonometry, EXT1 T2 SM-Bank 2

Using: $\cos 2A = 2 \cos^2 A - 1$

$$2 \cos^2 \frac{\pi}{8} - 1 = \cos \frac{\pi}{4}$$

$$2 \cos^2 \frac{\pi}{8} = \frac{1}{\sqrt{2}} + 1$$

$$\cos^2 \frac{\pi}{8} = \frac{1 + \sqrt{2}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

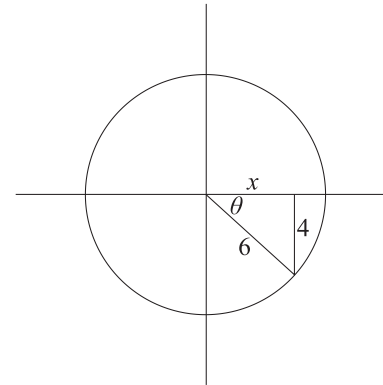
$$= \frac{\sqrt{2} + 2}{4}$$

$$\therefore \cos \frac{\pi}{8} = \sqrt{\frac{\sqrt{2} + 2}{4}}$$

$$= \frac{\sqrt{\sqrt{2} + 2}}{2}$$

24. Trigonometry, EXT1 T2 EQ-Bank 5

Consider the angle graphically:



COMMENT: Pay careful attention to the range of θ .

Since $\sin \theta$ is negative $\Rightarrow$ 4th quadrant
 $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Using Pythagoras,

$$x^2 = 6^2 - 4^2$$

$$x = \sqrt{20} = 2\sqrt{5}$$

$$\therefore \cos \theta = \frac{2\sqrt{5}}{6}$$

$$= \frac{\sqrt{5}}{3}$$

25. Trigonometry, EXT1 T2 EQ-Bank 7

$$\begin{aligned}
 \cos\left(\frac{11\pi}{12}\right) &= \cos\left(\frac{2\pi}{3} + \frac{\pi}{4}\right) \\
 &= \cos\left(\frac{2\pi}{3}\right) \cdot \cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{2\pi}{3}\right) \cdot \sin\left(\frac{\pi}{4}\right) \\
 &= \cos\left(\pi - \frac{\pi}{3}\right) \cdot \frac{1}{\sqrt{2}} - \sin\left(\pi - \frac{\pi}{3}\right) \cdot \frac{1}{\sqrt{2}} \\
 &= -\frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \\
 &= -\frac{(1 + \sqrt{3})}{2\sqrt{2}} \\
 &= -\frac{(\sqrt{2} + \sqrt{6})}{4}
 \end{aligned}$$

26. Trigonometry, EXT1 T3 SM-Bank 11

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$\cos \theta \cos \phi + \sin \theta \sin \phi = \frac{3}{5} \dots (1)$$

$$\frac{\sin \theta \sin \phi}{\cos \theta \cos \phi} = 2$$

$$\sin \theta \sin \phi = 2 \cos \theta \cos \phi \dots (2)$$

Substitute (2) into (1):

$$\cos \theta \cos \phi + 2 \cos \theta \cos \phi = \frac{3}{5}$$

$$3 \cos \theta \cos \phi = \frac{3}{5}$$

$$\cos \theta \cos \phi = \frac{1}{5}$$

Substitute $\cos \theta \cos \phi = \frac{1}{5}$ into (1):

$$\frac{1}{5} + \sin \theta \sin \phi = \frac{3}{5}$$

$$\sin \theta \sin \phi = \frac{2}{5}$$

$$\therefore \cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$= \frac{1}{5} - \frac{2}{5}$$

$$= -\frac{1}{5}$$

27. Trigonometry, EXT1 T3 SM-Bank 12

$$2 \sin x \cos x = \sin x$$

$$\sin x(2 \cos x - 1) = 0$$

$$\sin x = 0, \cos x = \frac{1}{2}$$

$$\text{If } \sin x = 0 \Rightarrow x = 0, \pi, 2\pi$$

$$\text{If } \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

MARKER'S COMMENT: Many students expanded $\sin(2x)$ and then cancelled $\sin x$ on both sides which lost a set of solutions!

28. Trigonometry, EXT1 T3 SM-Bank 8

$$x \sin(x) \sec(2x) = \frac{x \sin(x)}{\cos(2x)} = 0$$

Find x that satisfies:

$$x = 0 \text{ and } \cos(2x) \neq 0 \Rightarrow x = 0$$

or,

$$\sin(x) = 0 \text{ and } \cos(2x) \neq 0 \Rightarrow x = \pi, 2\pi$$

$$\therefore x = 0, \pi \text{ or } 2\pi$$

29. Trigonometry, EXT1 T3 2020 HSC 11d

$$A \sin(x + \alpha) = \sqrt{3} \sin x + 3 \cos x$$

$$A \sin x \cos \alpha + A \cos x \sin \alpha = \sqrt{3} \sin x + 3 \cos x$$

Equating co-efficients:

$$\Rightarrow A \cos \alpha = \sqrt{3}$$

$$\Rightarrow A \sin \alpha = 3$$

$$A^2 = (\sqrt{3})^2 + 3^2 = 12$$

$$\therefore A = \sqrt{12}$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{3}{\sqrt{3}}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\sqrt{3} \sin x + 3 \cos x = \sqrt{3}$$

$$\sqrt{12} \sin\left(x + \frac{\pi}{3}\right) = \sqrt{3}$$

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$\therefore x = \frac{\pi}{2}, \frac{11\pi}{6} \quad (0 \leq x \leq 2\pi)$$

30. Trigonometry, EXT1 T3 2021 HSC 11g

$$2\sin^3 x + 2\sin^2 x - \sin x - 1 = 0$$

$$2\sin^2 x(\sin x + 1) - (\sin x + 1) = 0$$

$$(2\sin^2 x - 1)(\sin x + 1) = 0$$

$$2\sin^2 x = 1 \quad \sin x = -1$$

$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

31. Trigonometry, EXT1 T3 EQ-Bank 3

$$\sqrt{3}\cos\theta = \sin(2\theta)$$

$$2\cos\theta\sin\theta - \sqrt{3}\cos\theta = 0$$

$$\cos\theta\left(\sin\theta - \frac{\sqrt{3}}{2}\right) = 0$$

$$\cos\theta = 0 \text{ or } \sin\theta = \frac{\sqrt{3}}{2}$$

COMMENT: Expressing 'all values' is specifically mentioned in Topic Guidance as an application of arithmetic sequences.

$$\theta = 90^\circ, 270^\circ \text{ or } \theta = 60^\circ, 120^\circ \quad (0^\circ \leq \theta \leq 360^\circ)$$

$$\therefore \theta = 90^\circ + m \times 180^\circ, 60^\circ + m \times 360^\circ \text{ or } 120^\circ + m \times 360^\circ$$

where m is an arbitrary integer.

32. Trigonometry, EXT1 T3 EQ-Bank 4

i. $f(t) = 6\sin 0.05t + 8\cos 0.05t$

$$\begin{aligned} A\sin(at + b) &= A\sin(0.05t + b) \\ &= A\sin(0.05t)\cos(b) + A\cos(0.05t)\sin(b) \end{aligned}$$

$$\Rightarrow A\cos(b) = 6, A\sin(b) = 8$$

$$A^2 = 6^2 + 8^2$$

$$A = 10$$

$$\Rightarrow 10\cos(b) = 6$$

$$\cos(b) = \frac{6}{10}$$

$$b = \cos^{-1} 0.6 \approx 0.927 \text{ radians}$$

$$\therefore f(t) = 10\sin(0.05t + 0.927)$$

ii. Max occurs at $\sin(0.05t + 0.927) = \sin \frac{\pi}{2}$

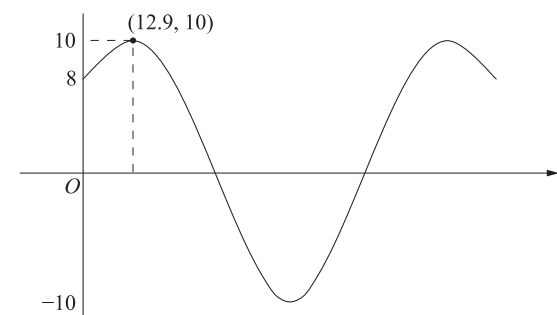
$$0.05t + 0.927 = \frac{\pi}{2}$$

$$0.05t = 0.643\dots$$

$$\therefore t = 12.87$$

$$= 12.9 \text{ (to 1 d.p.)}$$

iii.



33. Trigonometry, EXT1 T3 SM-Bank 2

$$\text{LHS} = \cos(2x + x)$$

$$= \cos 2x \cos x - \sin 2x \sin x$$

$$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin x \cos x \sin x$$

$$= \cos^3 x - \sin^2 x \cos x - 2 \sin^2 x \cos x$$

$$= \cos^3 x - (1 - \cos^2 x) \cos x - 2(1 - \cos^2 x) \cos x$$

$$= \cos^3 x - \cos x + \cos^3 x - 2 \cos x + 2 \cos^3 x$$

$$= 4 \cos^3 x - 3 \cos x$$

COMMENT: High achieving students know the 3 variants of $\cos 2x$ back to front. Here, $\cos 2x = \cos^2 x - \sin^2 x$ breaks the back of this problem.

34. Trigonometry, EXT1 T2 2021 HSC 13d

$$\begin{aligned} \text{i. } \frac{\sin A + \sin C}{\cos A + \cos C} &= \frac{\sin(B-d) + \sin(B+d)}{\cos(B-d) + \cos(B+d)} \\ &= \frac{2 \sin B \cos d}{2 \cos B \cos d} \\ &= \frac{\sin B}{\cos B} = \tan B = \text{RHS} \end{aligned}$$

$$\text{ii. Let } A = \frac{5\theta}{7}, C = \frac{6\theta}{7}$$

$$B = \frac{A+C}{2}$$

$$= \frac{1}{2} \left(\frac{5\theta}{7} + \frac{6\theta}{7} \right) = \frac{11\theta}{14}$$

$$\tan \frac{11\theta}{14} = \sqrt{3}$$

$$\frac{11\theta}{14} = \frac{\pi}{3}, \frac{4\pi}{3}$$

$$\therefore \theta = \frac{14\pi}{33}, \frac{56\pi}{33}$$

35. Trigonometry, EXT1 T3 2010 HSC 4b

$$\text{i. } 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right)$$

$$= 2 \cos \theta + 2 \left(\cos \theta \cos \left(\frac{\pi}{3} \right) - \sin \theta \sin \left(\frac{\pi}{3} \right) \right)$$

$$= 2 \cos \theta + 2 \cos \theta \times \frac{1}{2} - 2 \sin \theta \times \frac{\sqrt{3}}{2}$$

$$= 2 \cos \theta + \cos \theta - \sqrt{3} \sin \theta$$

$$= 3 \cos \theta - \sqrt{3} \sin \theta$$

$$R \cos(\theta + \alpha) = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$R \cos \alpha = 3 \quad R \sin \alpha = \sqrt{3}$$

$$\cos \alpha = \frac{3}{R} \quad \sin \alpha = \frac{\sqrt{3}}{R}$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \quad (0 < \alpha < \frac{\pi}{2})$$

$$R^2 = 3^2 + (\sqrt{3})^2$$

$$= 9 + 3$$

$$R = \sqrt{12} = 2\sqrt{3}$$

$$\therefore 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right)$$

$$\text{ii. } 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 3$$

$$2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right) = 3$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$$

♦ Mean mark part (ii) 49%
MARKER'S COMMENT: Many students did not check their answers against the stated domain for θ .

Since $\cos$ is positive in 1st and 4th quadrants,

$$\theta + \frac{\pi}{6} = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\therefore \theta = \frac{5\pi}{3} \quad (0 < \theta < 2\pi)$$

36. Trigonometry, EXT1 T3 EQ-Bank 1

$$\begin{aligned} \text{i. } \sin x + \sin 3x &= \sin x + \sin 2x \cos x + \cos 2x \sin x \\ &= \sin x + 2 \sin x \cos^2 x + (\cos^2 x - \sin^2 x) \sin x \\ &= \sin x + 2 \sin x \cos^2 x + \cos^2 x \sin x - \sin^3 x \\ &= \sin x + 3 \sin x (\cos^2 x) - \sin^3 x \\ &= \sin x + 3 \sin x - 3 \sin^3 x - \sin^3 x \\ &= 4 \sin x (1 - \sin^2 x) \\ &= 4 \sin x \cos^2 x \\ &= 2 \sin 2x \cos x \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{ii. } \sin x + \sin 2x + \sin 3x &= 0 \\ \sin 2x + 2 \sin 2x \cos x &= 0 \\ \sin 2x (1 + 2 \cos x) &= 0 \end{aligned}$$

If $\sin 2x = 0$:

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

If $\cos x = -\frac{1}{2}$:

$$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

37. Trigonometry, EXT1 T3 EQ-Bank 2

$$\text{i. } f(t) = \sqrt{5} \sin 0.2t + 2 \cos 0.2t$$

$$\begin{aligned} R \sin(nt - \alpha) &= R \sin(0.2t - \alpha) \\ &= R \sin 0.2t \cos \alpha - R \cos 0.2t \sin \alpha \end{aligned}$$

$$\Rightarrow R \cos \alpha = \sqrt{5}, \quad R \sin \alpha = -2$$

$$R^2 = (\sqrt{5})^2 + (-2)^2 = 9$$

$$R = 3$$

$$\cos \alpha = \frac{\sqrt{5}}{3}, \sin \alpha = -\frac{2}{3}$$

$\Rightarrow \alpha$ is in 4th quadrant

$$\text{Base angle} = \cos^{-1}\left(\frac{\sqrt{5}}{3}\right) = 0.7297$$

$$\therefore \alpha = 2\pi - 0.7297$$

$$= 5.553\dots$$

$$\therefore f(t) = 3 \sin(0.2t - 5.553)$$

$$\text{ii. Max value occurs when } \sin(0.2t - 5.553) = 1$$

$$0.2t - 5.553 = \frac{\pi}{2}$$

$$0.2t = 7.124\dots$$

$$t = 35.62\dots$$

Test if $t > 0$ for:

$$0.2t - 5.553 = -\frac{3\pi}{2}$$

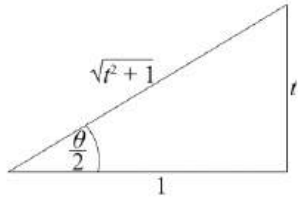
$$0.2t = 0.8406$$

$$t = 4.20\dots$$

$$\therefore \text{Time of 1st maximum: } t = 4.2$$

38. Trigonometry, EXT1 T3 2005 HSC 4b

Show $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$



$$\tan \frac{\theta}{2} = t$$

$$\Rightarrow \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = \frac{2t}{t^2 + 1}$$

$$\Rightarrow \cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \frac{1 - t^2}{t^2 + 1}$$

$$\Rightarrow \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2t}{1 - t^2}$$

$$\begin{aligned} \text{LHS} &= \frac{1}{\sin \theta} + \frac{1}{\tan \theta} \\ &= \frac{t^2 + 1}{2t} + \frac{1 - t^2}{2t} \\ &= \frac{t^2 + 1 + 1 - t^2}{2t} \\ &= \frac{1}{t} \\ &= \frac{1}{\tan \frac{\theta}{2}} \\ &= \cot \frac{\theta}{2} \\ &= \text{RHS} \dots \text{as required.} \end{aligned}$$

39. Trigonometry, EXT1 T3 2008 HSC 6b

Substitute $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
into $\sin 3\theta + \sin 2\theta = \sin \theta$,

$$3 \sin \theta - 4 \sin^3 \theta + \sin 2\theta = \sin \theta$$

$$2 \sin \theta - 4 \sin^3 \theta + 2 \sin \theta \cos \theta = 0$$

$$2 \sin \theta [1 - 2 \sin^2 \theta + \cos \theta] = 0$$

$$2 \sin \theta [1 - 2(1 - \cos^2 \theta) + \cos \theta] = 0$$

$$2 \sin \theta [1 - 2 + 2 \cos^2 \theta + \cos \theta] = 0$$

$$2 \sin \theta [2 \cos^2 \theta + \cos \theta - 1] = 0$$

$$2 \sin \theta (2 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$2 \sin \theta = 0$$

$$\Rightarrow \theta = 0, \pi, 2\pi$$

$$2 \cos \theta - 1 = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

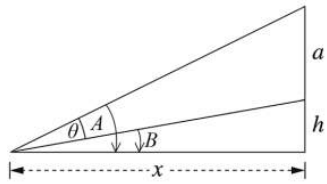
$$\cos \theta + 1 = 0$$

$$\cos \theta = -1$$

$$\Rightarrow \theta = \pi$$

$$\therefore \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi \quad (0 \leq \theta \leq 2\pi)$$

40. Trigonometry, EXT1 T3 SM-Bank 1



MARKER'S COMMENT: Answers that included a diagram and clearly labelled angles were generally successful.

$$\text{Show } \theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a+h)} \right]$$

$$\tan A = \frac{a+h}{x}$$

$$\tan B = \frac{h}{x}$$

$$\begin{aligned} \tan(A-B) &= \frac{\frac{a+h}{x} - \frac{h}{x}}{1 + \left(\frac{a+h}{x}\right)\left(\frac{h}{x}\right)} \times \frac{x^2}{x^2} \\ &= \frac{x(a+h) - xh}{x^2 + h(a+h)} \\ &= \frac{ax}{x^2 + h(a+h)} \end{aligned}$$

$$\text{Since } \theta = A - B$$

$$\theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a+h)} \right] \dots \text{as required.}$$

41. Trigonometry, EXT1 T3 2020 HSC 14b

i. Prove: $\sin^3 \theta - \frac{3}{4} \sin \theta + \frac{\sin(3\theta)}{4} = 0$

$$\begin{aligned} \text{LHS} &= \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{1}{4} (\sin 2\theta \cos \theta + \cos 2\theta \sin \theta) \\ &= \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{1}{4} (2 \sin \theta \cos^2 \theta + \sin \theta (1 - 2 \sin^2 \theta)) \\ &= \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{1}{4} (2 \sin \theta (1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta) \\ &= \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{1}{4} (2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta) \\ &= \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{3}{4} \sin \theta - \sin^3 \theta \\ &= 0 \end{aligned}$$

ii. Show $\sin(3\theta) = \frac{1}{2}$

Using part (i):

$$\begin{aligned} \frac{\sin(3\theta)}{4} &= \frac{3}{4} \sin \theta - \sin^3 \theta \\ \sin(3\theta) &= 3 \sin \theta - 4 \sin^3 \theta \dots (1) \end{aligned}$$

$$x^3 - 12x + 8 = 0$$

$$\text{Let } x = 4 \sin \theta$$

$$(4 \sin \theta)^3 - 12(4 \sin \theta) + 8 = 0$$

$$64 \sin^3 \theta - 48 \sin \theta = 0$$

$$\underbrace{-16(3 \sin \theta - 4 \sin^3 \theta)}_{\text{see (1) above}} = -8$$

$$-16 \sin(3\theta) = -8$$

$$\sin(3\theta) = \frac{1}{2}$$

♦♦♦ Mean mark part (iii) 21%.

iii. Prove: $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$

Solutions to $x^3 - 12x + 8 = 0$ are

$$x = 4 \sin \theta \text{ where } \sin(3\theta) = \frac{1}{2}$$

When $\sin 3\theta = \frac{1}{2}$,

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \dots$$

$$\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \dots$$

$\therefore$ Solutions

$$x = 4 \sin \frac{\pi}{18} \quad \left(= 4 \sin \frac{17\pi}{18} \right)$$

$$x = 4 \sin \frac{5\pi}{18} \quad \left(= 4 \sin \frac{13\pi}{18} \right)$$

$$x = 4 \sin \frac{25\pi}{18} \quad \left(= 4 \sin \frac{29\pi}{18} \right)$$

If roots of $x^3 - 12x + 8 = 0$ are α, β, γ :

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -12$$

$$\left(4 \sin \frac{\pi}{18}\right)^2 + \left(4 \sin \frac{5\pi}{18}\right)^2 + \left(4 \sin \frac{25\pi}{18}\right)^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$16 \left(\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} \right) = 0 - 2(-12)$$

$$\therefore \frac{\sin^2 \pi}{18} + \sin^2 \frac{5\pi}{18} + \frac{\sin^2(25\pi)}{18} = \frac{24}{16} = \frac{3}{2}$$

42. Calculus, EXT1 C2 2019 HSC 14c

i. $y_1 = \sin x$

$$\frac{dy_1}{dx} = \cos x$$

$$y_2 = \sin(x - \alpha) + k$$

$$\frac{dy_2}{dx} = \cos(x - \alpha)$$

At $x = x_0$, tangent is common

$$\therefore \cos x_0 = \cos(x_0 - \alpha)$$

♦ Mean mark part (i) 47%.

ii. x_0 is in 1st quadrant (given)

Using part (i):

$$\cos x_0 = \cos(x_0 - \alpha) > 0$$

$$\Rightarrow x_0 - \alpha \text{ is in 4th quadrant } (0 < \alpha < \pi)$$

Since sin is positive in 1st quadrant and

negative in 4th quadrant

$$\Rightarrow \sin x_0 = -\sin(x_0 - \alpha)$$

♦♦♦ Mean mark part (ii) 19%.

iii. When $x = x_0$,

$$y_1 = \sin x_0$$

$$y_2 = \sin(x_0 - \alpha) + k$$

$$\sin x_0 = \sin(x_0 - \alpha) + k$$

$$= -\sin x_0 + k$$

$$k = 2 \sin x_0$$

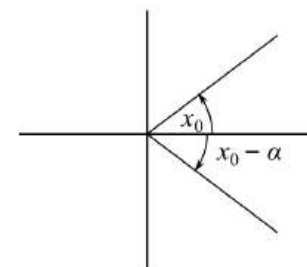
Since $\cos x_0 = \cos(x_0 - \alpha)$

$$x_0 = -(x_0 - \alpha)$$

$$2x_0 = \alpha$$

$$x_0 = \frac{\alpha}{2}$$

$$\therefore k = 2 \sin \frac{\alpha}{2}$$



♦♦ Mean mark part (iii) 21%.